How Costly Are Cartels?

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We study the cost of cartels in an oligopoly model with heterogeneous firms, endogenous markups, and collusion. Cartels can amplify or dampen misallocation, by charging supracompetitive markups and reallocating demand towards non-colluding firms. Using French micro data, our calibrated model suggests that cartels operating in the most productive industries may decrease aggregate productivity by up to 0.4% and welfare by up to 0.7%. We also find that standard competitive oligopoly models may understate the cost of markups by about 20%.
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Cartels are secret and illegal agreements that, by definition, are hard to observe. However, evidence suggests that "some forms of collusion are likely to be prevalent in many industries" (Asker and Nocke, 2021).[1] If industries are cartelized, how costly might collusion be?

The lack of quantitative work on cartels in macroeconomics is unlikely to be due to lack of importance or interest.[2] The variety of cartel arrangements requires a macroeconomic framework rich enough to accommodate collusive firms of varying sizes, markups, and overcharges alongside firms who behave competitively. The goal of this paper is to propose a flexible yet parsimonious approach

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[1] From the infamous 1920s Phoebus cartel, a landmark case of planned obsolescence with light bulbs engineered to be shorter-lived, to vitamin or car glass cartels, cooperation between competing firms on the prices charged to customers is typically forbidden by antitrust authorities. We focus on horizontal cartels that affect prices.

[2] From a theoretical standpoint, cartels may have little effect on the economy if competitive forces —incentives to defect from the collusive agreement (Stigler, 1964)—are strong enough to make cartels unstable and irremediably short-lived. In contrast, the recent empirical literature has shown that cartels are long-lasting (Levenstein and Suslow, 2006; Hyytinen, Steen and Toivanen, 2018).

to quantify the macroeconomic cost of cartels. As a result, our model allows us to integrate the empirical cartel literature with recent macroeconomic studies on the productivity and welfare costs of markups, and, in particular, to disentangle markups stemming from the unilateral exercise of market power (Edmond, Midrigan and Xu, 2023; Baqaee and Farhi, 2020) from overcharges arising from collusion between firms.

Specifically, we extend Atkeson and Burstein (2008)'s static heterogenous-firm model with oligopolistic competition by introducing collusive behaviors, which we microfound using the tractable cross-ownership framework of O'brien and Salop (1999). Heterogenous markups arise endogenously in the model as more productive firms have a large market share, thereby facing a lower demand elasticity and allowing them to charge higher markups in equilibrium. Cartel members deviate from own-profit maximization, as they internalize some of their impact on the profits of other cartel members. In equilibrium, cartel members' demand elasticities thus depend on their own market shares as well as the market share of other cartel members weighted by an intensity of collusion parameter. As a result, cartel members face a lower demand elasticity and charge supracompetitive markups, i.e., overcharges,[3] which affect markup dispersion and the level of the aggregate markup. Importantly, the effect of cartels on sectoral productivity is ambiguous: while supracompetitive markups entail an output drop of colluding firms, they also trigger market share reallocations towards non-colluding firms. Depending on the composition of cartels, sectoral productivity may increase or decrease. The effect on aggregate productivity further depends on which industries are cartelized. Intuitively, since the economy features significant cross-sectoral heterogeneity (for instance, the top 4 sectors in France account for 14% of overall French sales), collusion in large sectors may have substantial aggregate effects.

We calibrate our model to French data as follows. First, we assign cartels to the top decile in terms of sectoral productivity. This helps us reproduce the positive correlation between an industry's market share and the presence of a cartel in that industry, as well as the average share of cartelized industries found in the data. Additionally, this allows us to assess the impact of large cartelized sectors, providing an upper bound. Second, we assume that cartels in cartelized sectors are made up of top producers. This is motivated by our empirical evidence that cartel members tend to be much larger in terms of observable firm characteristics than non-cartel members, even within narrowly defined industries. We further show that cartels are made up of relatively homogeneous firms in terms of productivity and sales. While these results are to be interpreted with caution given that colluding firms have been detected by the competition authority, they are nevertheless consistent with recent work on cartel formation choice showing that cartels can be expected to be made up of the largest firms (Bos and Harrington,

[3]Cartels differ from horizontal mergers in which cost synergies might increase the productivity of the firms involved in the merger and dominate the increase in market power, as in the classical trade-off model of Williamson (1968) or in the dynamic setting of Moreau and Üslü (2025).

2010, 2015). Third, we require the model to match moments such as the level of the aggregate markup, the amount of sales concentration within and across sectors, the relationship between market shares and markups, the median number of operating firms and the median number of cartel members. Finally, the intensity of collusion parameter is estimated directly from the micro data. It is determined by the relative importance of the cartel market share in influencing firm-level inverse markups. This parameter pins down the overcharge induced by cartels and allows us to conduct policy-relevant counterfactual exercises —as we drive the amount of overcharge from positive values to zero. As we compute the cost of cartels relative to a counterfactual competitive economy, our framework complements the methodologies of antitrust authorities across OECD countries to assess the damage cartels inflict on the economy.[4]

To assess the impact of cartels operating in top industries on aggregate productivity and welfare, we study how these variables change as the intensity of collusion parameter is reduced to zero. We find that aggregate TFP would be 0.2% higher if there were no cartels. Intuitively, because cartel members are the top firms in their industry, breaking them down reduces markups and reallocates demand towards these large producers. This decreases markup dispersion and increases aggregate productivity. Second, our framework also has implications in terms of distance to the efficient allocation, where relative prices are aligned with relative marginal costs. We find that eliminating cartels —thereby reducing the extra amount of markup dispersion —would bring the economy 18% closer to the efficient allocation. This suggests that eliminating cartels can be an effective way of improving allocative efficiency. Third, the decrease in the *level* of markups leads to a drop in the aggregate markup, which also has welfare implications. We find that eliminating cartels would lead to a consumption-equivalent welfare gain of about 0.5%. Finally, we find that the productivity (welfare) cost of markups computed with respect to the efficient allocation is 20% (4%) higher in our calibrated oligopoly model with collusion than in a calibrated competitive oligopoly model. In other words, although both types of models are calibrated to the same data, our oligopoly version with collusion generates relatively more misallocation to start with, thereby increasing the cost of markups highlighted in the recent literature (Edmond, Midrigan and Xu, 2023).

These numbers challenge the received wisdom that the economic cost of distortions to competition might be low, as they are higher than the estimate provided by Harberger (1954). We show that our model can reproduce Harberger's estimate when we instead assume a sectoral version of the model with no markup dispersion across firms within industries, or when the demand elasticities are close to unity —as he assumed using sectoral data. Intuitively, this is because larger elasticities of substitution generate more markup dispersion, as does relying on

[4]Antitrust authorities typically assume a cartel overcharge equal to 10% (OECD, 2014), and, in addition, make other assumptions on the duration of the anticompetitive practice and the extent to which sales on the relevant market were affected.

disaggregated data, as also recently shown by Baqaee and Farhi (2020).

Breaking down cartels would also increase competition through a second channel: umbrella pricing. Indeed, the presence of a cartel allows non-cartel members to increase their markups and prices as the prices of cartel members serve as an umbrella, as recognized for instance by the Court of Justice of the European Union.[5] We find that this umbrella pricing effect dampens the aggregate gains to productivity and welfare but that the effect is quantitatively small. We also study the welfare gains of competition policy at the intensive margin, i.e., cartel members respond to more vigilant antitrust scrutiny by reducing their collusion intensities. We find that the intensive margin of cartels is important too. Halving the collusion intensity parameter still generates gains to aggregate productivity and welfare equal to 0.13% and 0.22%, respectively.

Our results are robust to matching a lower aggregate markup level to account for uncertainty regarding this target, to allowing firms to compete à la Bertrand, and to allowing the intensity of collusion to differ across cartels. Importantly, we find that the cost of collusion is about twice as large when cartels operate in the top decile of industries instead of the eighth decile, or when the intensity of collusion is initially twice as high. We also show that cartels operating in industries similar to those in which they operated in two extreme years (in terms of the share of cartelized industries) lead to average aggregate productivity and welfare losses of 0.1%. Since these are based on convicted cartels, one can consider these numbers, which are close to Harberger (1954)'s original estimate, as being a lower bound.

The cost of cartels that we estimate could be affected if markup dispersion arose primarily from other consideration than size-related market power, as in our model, such as geographical differentiation. Furthermore, our static model abstracts from dynamic considerations such as higher entry of firms stimulated by larger expected profits or innovation investments to escape competition. With this in mind, our quantitative results nevertheless suggest that the static cost of cartels is important, especially if cartels operate in the largest industries.

Related Literature. — Misallocation of factors of production is an important source of productivity loss (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). We analyze markup dispersion as a source of misallocation, as in Edmond, Midrigan and Xu (2023) and Baqaee and Farhi (2020), but focus on a specific type of competition distortion. In particular, our main exercise is directly policy-relevant and consists in quantifying the additional misallocation costs arising from cartels compared to the oligopolistic-competitive allocation, which remains inefficient as markup dispersion is still present. While it might be hard to implement policies that fully eliminate markup dispersion, eliminating the extra dispersion caused by the presence of cartels is arguably more easily attainable through com-

[5] Case C-557/12 Kone ("Kone").

petition policy.[6] Moreover, we microfound our model, show how competitive and collusive models may yield different results and that there may be important gains at the intensive margin, in contrast to Edmond, Midrigan and Xu (2015) who consider an extension of their framework with collusion when looking at the gains from trade in Taiwan. We abstract, however, from free entry. For these reasons, we view our contribution and that of Edmond, Midrigan and Xu (2015, 2023) and Baqaee and Farhi (2020) as complementary.

Our paper is also related to recent work that quantifies the cost of non-competitive behaviors. Our microfoundation of collusion builds on cross-ownership models à la O'brien and Salop (1999). These models were recently revived by Azar, Schmalz and Tecu (2018) and are featured in contemporary work such as that by Ederer and Pellegrino (2021). Our approach differs by embedding collusion within the oligopoly model of Atkeson and Burstein (2008). This enables us to focus on the aggregate productivity effect of collusion and show how standard competitive models may understate misallocation. In related and important work, Brooks, Kaboski and Li (2021) develop a screen for identifying noncompetitive behaviors and apply it to China's manufacturing clusters. They find that while firms in clusters charge higher markups, the welfare impact can be positive due to a decline in markup dispersion both within the cluster and in the industry as a whole. In contrast, we find that collusive behavior in our setting decreases welfare. French cartels often consist of the *most* productive firms, which reduces sectoral productivity by reallocating demand to less efficient firms.[7] One contribution of this paper is to show that the welfare implications of collusion are ambiguous and depend on the composition of cartels (for sectoral productivity) and on the share and size of cartelized industries (for aggregate productivity).

Our work also relates to recent influential papers that document the rise of markups (De Loecker and Eeckhout, 2018; De Loecker, Eeckhout and Unger, 2020; De Loecker and Eeckhout, 2025) and link changes in market concentration to changes in the labor share (Autor et al., 2020; De Loecker, Eeckhout and Mongey, 2021). Although we focus on quantifying the gains from breaking down cartels, our framework also has implications in terms of the relationship between competition and market power over time and across markets. Gutiérrez and Philippon (2018) argue that laxer antitrust enforcement in the US is behind the larger increase in concentration observed in the US compared to Europe. If lax antitrust enforcement allows cartels to develop and prosper, this would reduce competition within sectors, increase the market power of all firms, thereby driving up the aggregate markup and depressing the aggregate labor share. Our findings are in line with those of Vaziri (2022) who finds that stronger antitrust policies that help fight aggressive strategies (i.e., aggressive pricing, killer acquisitions

[6]Our results further provide a mechanism why measures that increase competition improve productivity as Buccirossi et al. (2013) document for 22 industries in twelve OECD countries over 1995 to 2005.

[7]On the other hand, syndicates may not be only composed of top producers as in Brooks, Kaboski and Li (2021), which could yield clusters to partly reallocate demand towards more efficient firms.

etc.) increase productivity growth and welfare.

Our work also builds on recent theoretical and empirical advances on the economics of cartels. Bos and Harrington (2010) study cartel formation with heterogeneous firms. They show that larger firms have a strong incentive to form a cartel when they are patient enough, and that smaller firms can increase their prices as the larger firms' prices serve as an umbrella. We provide evidence that discovered cartel members are more productive and are larger than non-cartel members in their industry. The empirical study of cartels and their impact on productivity is limited by the fact that secret agreements are, by definition, hard to observe.[8] It is possible, however, to focus on specific cartels operating in particular industries. Bridgman et al. (2015) estimate that the New Deal sugar cartel tremendously decreased productivity through reallocation of production towards low productivity firms in the beet and cane industries. Asker, Collard-Wexler and De Loecker (2019) focus on the oil industry and quantify the role of market power in generating misallocation. Our paper instead connects the cartel and macroeconomics literature by looking at the productivity and welfare implications of cartels from a macroeconomic perspective.

The paper proceeds as follows. Section I details our data. Section II introduces the model. Section III provides more information on the quantification of the model. Our results, robustness and sensitivity experiments are presented in Section IV. In Section V, we discuss some of our assumptions and estimation strategy. Section VI concludes. Proofs and supplemental materials are in the online Appendix.

I. Data

In this section, we describe our data and certain characteristics of cartels.

A. Antitrust Decisions

The French Competition Authority (ADLC) is primarily in charge of investigating and fining companies operating on the French market that are found guilty of engaging in any form of anticompetitive practice, i.e., abuse of dominant position, collusion or predatory pricing.[9] We focus on collusion between firms so that anticompetitive practices will refer to collusion hereafter. Collusive behaviors might involve firms trading information on their prices, imposing standard form contracts, enforcing barriers to entry, imposing exclusive or selective distribution agreements, market sharing, purposely stepping down from calls for bids, or a combination of the above.

[8] Levenstein and Suslow (2006) survey the literature on cartels. Most papers study the impact of cartels on prices or the determinants of cartels' success (Levenstein and Suslow, 2011). Some papers instead study the impact of cartels on welfare, such as Röller and Steen (2006) in the context of the Norwegian cement industry, or output and wages in the context of Mexico (Reed et al., 2022).

[9] The variety of possible collusive arrangements, which include price fixing, production limitations or market sharing, are recognized by Article 101 of the Treaty on the Functioning of the European Union.

We assemble a firm-level dataset on French cartels over the period 1994-2007, using the written reports of all the antitrust decisions taken by the ADLC over the last decades. Our database summarizes information contained in the investigation and decisions files published in French on the ADLC website. Crucially, the PDF files contain the name of the firms that are fined for engaging in anticompetitive practices.[10] We also retrieve information on the amount of the fine, the type of anticompetitive practice, the duration of the practice, the cause of breakup, the year the verdict is returned and the starting year of the investigation. We then use the companies' names and sales to recover their unique national identification code ("SIREN" code) given by the French National Institute of Statistics and Economic Studies (Insee). This allows us to match our database to other firm-level production datasets. More details on the construction of the database can be found in online Appendix A.[11]

Because our analysis focuses on a single country and because information on market shares of foreign firms on the French market cannot be recovered, international cartels are not included in our data. These cross-country cartels are usually investigated by the European Commission and its Directorate General for Competition (DG Comp), which deals with cases affecting multiple European member states. Given that these private international cartels are typically "the largest, most injurious, and most difficult to prosecute of all price-fixing violations" (Connor, 2020),[12] the estimates we provide based on national cartels will likely *underestimate* the impact of collusion. We eliminate from our dataset cases where single firms were fined for behaving anticompetitively. This is the case if firms abused their dominant position or are repeat offenders, for example. Our final dataset on cartels contains 174 cartels and over a thousand firms before merging, and over 800 firms after merging.

B. Administrative Data

We match our database on anti-competitive firms with firm-level data for France, using the firms' identification number. The datasets that we use (FICUS) contain the universe of French firms over the period 1994-2007 (Insee, 1994-2007*b*). These datasets contain the balance sheets and income statements of all French firms. We keep both large and smaller firms, which correspond to two different tax regimes: the Regime of Normal Real Profits (BRN) and the Simplified Regime for the Self-Employed (RSI), respectively. BRN contains firms with annual sales

[10] We use automatic textual analysis and manual checks to identify firms fined by the antitrust body and merge this information with the other datasets, excluding firms only notified by injunction, as they often appear later in our database.

[11] The number of decisions per year and the number of firms involved are reported in online Appendix Figure F1 and F2. Online Appendix Figure F3 displays a representative report from the ADLC (decision file 17d20). Online Appendix Figure F4 shows that information on the duration of cartel can also be found in these reports, as well as information on the type of infringement (online Appendix Figure F5).

[12] As Connor (2020) explains, "private cartels are those that are not protected by government sovereignty or by international treaties, like the Organization of Oil Exporting Countries (OPEC)".

above 763K euros (230K euros for services) whereas smaller firms included in RSI sell at least 76.3K euros (but less than 763K euros) a year and more than 27K euros for services. However, BRN is the most relevant data source given that in 2003, BRN firms' sales share in total sales was 94.3% and is constant over time. This data source has been used in previous studies, for instance in Di Giovanni, Levchenko and Mejean (2014), and we refer to their paper for more information. Importantly, these exhaustive databases allow us to recover a firm's market share and other variables we use in our empirical analysis. More information on the variables we use can be found in online Appendix A4. Summary statistics of our main variables are reported in online Appendix Table E1 and online Appendix Table E3.

C. Characteristics of Cartels

Cartel Duration, Size, and Activities. — The average duration of a cartel is about 4.5 years (see online Appendix Table E2), which is close to the average duration reported in Monnier-Schlumberger and Hutin (2016) who report an average duration of five years for their sample of discovered French cartels observed over the period 2003-2015. This also matches the average duration of cartels summarized in Levenstein and Suslow (2006) for a wide range of studies. Our median duration is about 3 years, which is also consistent with what Monnier-Schlumberger and Hutin (2016) report (3.8 years).[13] The average number of firms per cartel is 6 and the median is 4. While there are extremely large cartels made up of more than 70 firms, this is not the norm as the standard deviation is equal to 7. Combe and Monnier (2012) report an average (median) number of firms per cartel of 7.7 (5), while Monnier-Schlumberger and Hutin (2016) report an average number of cartel members of 10. We further report a few statistics on the types of cartels. Most firms that are part of a cartel share confidential information, rig procurement auctions, and fix their prices. Communicating seems to be a pervasive feature of cartels. As Asker and Nocke (2021) argue, "across the heterogeneity of cartel forms, a relatively common feature, empirically, of coordinated activity that seems uncontroversially anti-competitive is communication". They also share their customers and their market shares, which has been found to be the type of practice that allows cartels to sustain their illegal activities for a long time (Combe and Monnier, 2012; Levenstein, 2006).

Cartelized Industries. — Cartels are prevalent in France over the period 1994-2007. Out of 53 2-digit sectors over the time period, cartels operated in 38 of them, confirming the finding that cartels affect intermediate good sectors, as well

[13] Combe and Monnier (2012) find that the average (median) duration of cartels is 7 (6) years for their sample of European cartels detected and fined by the European Commission over 1969-2009.

Table 1—Cartelized Industries are Larger

Dependent variable:	Industry market share	
	(1)	(2)
Cartelized industry	0.195	0.195
	(0.034)	(0.034)
Year FE	No	Yes
Observations	9,177	9,177
Adj. R^2	0.03	0.03

Notes: Standard errors clustered at the 4-digit industry level. The dependent variable is the sales share of a 4-digit industry. The independent variable is a dummy variable equal to one if a cartel operates in that industry in a given year.

as other sectors such as services (Monnier-Schlumberger and Hutin, 2016).[14] As shown in online Appendix Table E4, on average, close to 10% of 4-digit industries are cartelized over 1994-2007 and these cartelized industries account for 20% of overall French sales. Table 1 reports the correlation between the presence of a cartel in a 4-digit industry and the size of that industry (measured by its market share). Cartelized industries are larger than non-cartelized industries, even within 2-digit sectors (online Appendix Table E5).

Cartel Members. — We provide novel evidence that cartel members are, on average, larger than non-members,[15] and more homogeneous in terms of labor productivity and sales, even within narrowly defined industries. Table 2 shows that there exist important size differences between cartel members and non-members

[14]Levenstein and Suslow (2006) report cartels spanning the beer, bromine, cement, coal, diamonds, electrical equipment, ocean shipping, oil, parcel post, potash, railroad, rayon, steel, sugar and tea industries.

[15]There are two important caveats regarding the numbers we provide. First, unfortunately, the market share of each cartel member is seldom reported in the decision files. We circumvent this issue by calculating the market share of cartel members and the cartel market share using our administrative micro data. The market shares are defined at the 4-digit level —the highest level of disaggregation in our data. Second, our sample of cartels consists of *discovered* cartels which may not be representative of the latent population of cartels (Harrington Jr and Wei, 2017). Indeed, there might be a myriad of other cartels and colluding companies that go unnoticed —therefore classifying as competitive —while behaving differently from discovered firms. On the one hand, small undetected cartels might break down quickly because they are "bad" at colluding and do not use compensation schemes, for instance —which are typically found to be important in preventing cartel breakdowns (Levenstein and Suslow, 2011). In this case, our numbers would overestimate the size differences between anticompetitive and competitive firms. On the other hand, very large undetected cartel members might be able to go unnoticed because of their capacity to avoid detection and prosecution. This would lead us to underestimate the size differences between cartel members and competitive firms. Although it is not possible to assess the direction of the bias, the theoretical arguments highlighted in online Appendix B would point in the direction of a downward bias.

Table 2—Cartel Members Are Larger

Dependent variable:	Sales (ln) (1)	Market Share (2)	Employment (ln) (3)	Labor Productivity (ln) (4)
Cartel member	2.444	2.574	2.054	0.237
	(0.070)	(0.300)	(0.062)	(0.018)
Four-digit Industry × Year FE	Yes	Yes	Yes	Yes
Observations	8,312,229	8,312,229	8,312,229	8,312,229
Adj. R^2	0.353	0.189	0.220	0.185

Notes: Standard errors clustered at the firm level. The dependent variables are log sales, the firm's market share, log employment, and labor productivity, which is measured as value-added (deflated by 2-digit price indices) divided by the number of workers. The independent variable is a dummy variable that equals one if the firm is a cartel member. The periods during which firms collude have been excluded.

by regressing several firm-level observable characteristics on a dummy variable equal to one if firms are cartel members. We have excluded the period over which cartel members collude to focus on self-selection into colluding. Specifically, the table suggests that cartel members are about 12 times and 8 times larger than competitive firms in terms of sales and employment, have a market share higher by almost 3 percentage points, and have a 27% higher labor productivity. This also holds for markups (online Appendix Table E6).

Moreover, online Appendix Table E7 highlights the fact that productivity and sales differences across cartel members are smaller than those across competitive firms in narrowly defined industries.[16] Panel A reports different productivity distribution moments for non-cartel members in their 4-digit industries (columns 1 to 3) and for cartel members within their cartel (column 4 to 6) in 2007. The first three columns echo the findings of Syverson (2004), namely, that there are important productivity differences even within narrowly defined industries. For instance, column 1 indicates that the average within-industry interquartile range is about 0.72, which means that firms in the 75th percentile of an industry's productivity distribution are about twice as productive as firms in the 25th percentile. However, this ratio is only 1.6-to-1 across firms within their cartel. Similarly, the average 90-10 and 95-5 percentile productivity ratios across non-cartel members within industries are over 4 to 1 and 8 to 1, respectively. These numbers are 2 to one for cartel members. These findings further extend to sales as shown in Panel B. As a case study, online Appendix Table E8 illustrates this finding for the manufacture of plastic components for construction industry in which two cartels operated in 2007.

All in all, the fact that cartel members are larger than non-cartel firms within industries, and that cartelized industries tend to be economically larger than

[16]The fact that firms within a cartel are relatively homogeneous, lends empirical credence to the theoretical argument that large cost differences across cartel members might make collusion less easily sustainable (Ivaldi et al., 2007). This is because large firms might be better protected from retaliation in case they deviate from the collusive equilibrium and because large firms might gain relatively more by cheating. Miklos-Thal (2011) provides an interesting theoretical treatment of price fixing with asymmetric participants.

non-cartelized ones, is consistent with the idea that cartel members can shape sectoral and aggregate outcomes, in the spirit of Gaubert and Itskhoki (2021). This motivates the use of a framework with granular firms and cross-sectoral heterogeneity.

II. Model

We build a static, closed-economy, model in which heterogeneous firms choose their markups endogenously along the lines of Atkeson and Burstein (2008), and where cartels coexist with competitive firms. The model allows for both Cournot and Bertrand competition. The economy consists of a continuum of sectors in which a *finite* number of firms compete with each other. In equilibrium, firms' endogenous markups increase with their market share.

The most attractive feature of our framework is its tractability: it nests several modes of collusion depending on the value of a single parameter. Following Harrington Jr (2017)'s definition of collusive behavior,[17] collusion in our model affects the extent to which firms internalize the impact of their production and pricing decisions on the sectoral output and price level. We micro-found collusion by building on cross-ownership models (O'brien and Salop, 1999), which formally produce similar competition distortions.[18] The collusion intensity parameter κ in our model captures what is arguably the most important feature of cartels, cartel overcharges, which distort price schedules and affect aggregate productivity and welfare.

A. Environment

Market Structure. — The production side of the economy consists of a continuum of sectors indexed by $s \in [0, 1]$. Final consumption c is produced by a competitive firm that combines the outputs from all the sectors y_s with a CES technology with elasticity η:

$$c = \left[\int_0^1 y_s^{\frac{\eta-1}{\eta}} ds \right]^{\frac{\eta}{\eta-1}}. \tag{1}$$

The inverse demand function for each intermediate output from sector s is given by:

$$\frac{P_s}{P} = \left(\frac{y_s}{c} \right)^{-\frac{1}{\eta}}, \tag{2}$$

[17] "Collusion is when firms in a market *coordinate* their behavior for the purpose of producing a *supracompetitive outcome*" (Harrington Jr, 2017, p.1, emphasis in original).

[18] Gilo, Moshe and Spiegel (2006) and de Haas and Paha (2016) study how common ownership affects collusion.

where P, the price index for final consumption representing the "true cost of living", is a function of the sectoral prices:

$$P = \left[\int_0^1 P_s^{1-\eta} ds\right]^{\frac{1}{1-\eta}}. \tag{3}$$

Each sector is populated by a finite number of firms K_s indexed by k. Because each firm has a non-zero measure, it is therefore "large in the small but small in the large" (Neary, 2003), i.e., firms are small with respect to the economy but large in their own sector. The output of sector s is a composite of the firms' outputs, combined with a CES technology with elasticity parameter ρ:[19]

$$y_s = \left[\sum_{k=1}^{K_s} (q_{sk})^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}. \tag{4}$$

The inverse demand functions within each sector are given by:

$$\frac{P_{sk}}{P_s} = \left(\frac{q_{sk}}{y_s}\right)^{-\frac{1}{\rho}}, \tag{5}$$

where the price index P_s in sector s is given by

$$P_s = \left[\sum_{k=1}^{K_s} (P_{sk})^{1-\rho}\right]^{\frac{1}{1-\rho}}. \tag{6}$$

We consider an industry s populated by K_s firms, of which a subset $\mathcal{C}_s$ forms a cartel, with $0 \leq |\mathcal{C}_s| \leq K_s$. For simplicity, we abstract from vertical arrangements and assume that firms form horizontal cartels that do not reach across industries. Moreover, we derive our main results under Cournot competition but our results are qualitatively robust to assuming Bertrand competition as shown in the online Appendix and in the robustness section.

Non-cartel Members. — With linear labor costs and heterogenous productivity z_{sk}, any competitive firm that does not belong to the cartel ($k \notin \mathcal{C}_s$) solves the following maximization problem:

$$\max_{q_{sk}} \left(P_{sk} - \frac{W}{z_{sk}}\right) q_{sk}, \tag{7}$$

[19] Goods are imperfect substitutes, $\rho < \infty$, and more substitutable within than between sectors, $1 < \eta < \rho$.

subject to the inverse demand function

$$\left(\frac{P_{sk}}{P}\right) = \left(\frac{q_{sk}}{y_s}\right)^{-\frac{1}{\rho}} \left(\frac{y_s}{c}\right)^{-\frac{1}{\eta}}.$$

Profit-maximization implies that the equilibrium price is a markup μ_{sk} over the marginal cost of production, where the markup is pinned down by the idiosyncratic demand elasticity ε_{sk} faced by the firm,

$$\begin{aligned} \mu_{sk} &= \frac{\varepsilon_{sk}(\omega_{sk})}{\varepsilon_{sk}(\omega_{sk}) - 1} \\ \varepsilon_{sk}(\omega_{sk}) &= \left[\frac{1}{\rho} + \left(\frac{1}{\eta} - \frac{1}{\rho}\right)\omega_{sk}\right]^{-1}, \end{aligned} \tag{8}$$

where $\omega_{sk} := \frac{P_{sk}q_{sk}}{\sum_{j=1}^{K} P_{sj}q_{sj}}$ is the sectoral revenue share of firm i. Firms with larger equilibrium market shares have more market power and therefore charge higher markups. In particular, the CES demand structure implies that the demand elasticity that each firm faces in equilibrium is a harmonic weighted average of the within and between-elasticities.

Cartel Members. — Collusion distorts firms' profit incentives. Instead of maximizing their own profits independently, members of the same cartel internalize that their decision impacts the other cartel members. The distorted objective function for cartel member k takes the following form:

$$\pi_k^{\mathcal{C}} \propto \pi_k + \sum_{j \in \mathcal{C} \setminus \{k\}} \kappa \cdot \pi_j, \tag{9}$$

where π_k corresponds to the profits realized by firm k and the parameter κ captures the intensity of collusion.[20] This flexible formulation yields straightforward analytical derivations of collusive behaviors of various intensities and sizes. It can also easily be adapted to allow for more complex collusion arrangements with varying collusion intensities within the cartel (see online Appendix D for the more general framework). In addition, it is consistent with micro-foundations that could cover side payments or ringleaders exerting control over the production decisions of other, often smaller, cartel members.[21]

[20] Rescale the right hand side by a factor $\frac{1}{1+\kappa(|\mathcal{C}_s|-1)}$ for the exact expression, simplified for exposition. The first-order conditions are unaffected.

[21] To see this, let β_{jl} denote the share of firm j's operational profits promised as a side-payment to firm l. The financial profits accruing to firm l's shareholders then correspond to the portfolio $\pi^l = \sum_j \beta_{jl}\pi_j$. If firm k is under the influence of other members of the cartel, then the distorted objective function of its managers becomes $\tilde{\pi}_k = \sum_{l \in \mathcal{C}} \gamma_{kl}\pi^l = \sum_l \gamma_{kl} \sum_j \beta_{jl}\pi_j$ where γ_{kl} denotes firm l's control over firm k's

Cartel members therefore solve the following maximization problem:

$$\max_{q_{sk}} \left[\Big(P_{sk} - \frac{W}{z_{sk}} \Big) q_{sk} + \kappa \sum_{j \in \mathcal{C} \setminus \{k\}} \Big(P_{sj} - \frac{W}{z_{sj}} \Big) q_{sj} \right], \quad \forall k \in \mathcal{C}_s \tag{10}$$

subject to

$$\left(\frac{P_{sk}}{P} \right) = \left(\frac{q_{sk}}{y_s} \right)^{-\frac{1}{\rho}} \left(\frac{y_s}{c} \right)^{-\frac{1}{\eta}}.$$

Markups take a similar form as in equation (8) but collusion weakens competition, generating demand elasticities $\varepsilon^{\mathcal{C}}_{sk}$ for cartel members given by

$$\varepsilon^{\mathcal{C}}_{sk}(\omega_{sk}) = \left[\frac{1}{\rho} + \Big(\frac{1}{\eta} - \frac{1}{\rho} \Big) \Big(\omega_{sk} + \kappa \sum_{j \in \mathcal{C} \setminus \{k\}} \omega_{sj} \Big) \right]^{-1}. \tag{11}$$

One observation is noteworthy. Cartels do not affect fundamentals - firms' productivity - but distort cartel members' production decisions, as they partially internalize the effect of their own production decisions on other members' profits. This can be seen from equation (11) in which the market share of other cartel members lowers cartel members' own demand elasticity, allowing them to charge higher markups.

B. Types of Collusion

The extent to which colluding firms internalize part of the effect of their decision on the other cartel members' profits depends on the collusion intensity parameter κ. Importantly, our model nests several cases of interest.[22]

Benchmark Competitive Economy ($\kappa = 0$). — When the collusion parameter κ is set to zero, there are no cartels and the model boils down to a competitive Nash-Cournot model with heterogeneous firms (Atkeson and Burstein, 2008). In this case, firms' markups are given by equation (8), with more productive firms charging higher markups. This is the main counterfactual allocation we consider to compute the aggregate cost of cartels.

operational decisions, that forces firm k's managers to internalize the impact of their decisions on firm l's profits. For the distinction between ownership and control, see Shleifer and Vishny (1997). See O'brien and Salop (1999), Azar, Schmalz and Tecu (2018), Azar and Vives (2021), and Ederer and Pellegrino (2021) for related formulations in the common ownership literature.

[22] The micro-founded model in the online Appendix details configurations that support the cases detailed below, as well as other cases.

Cartels with Symmetric Collusion ($\kappa > 0$). — The second case we consider is that of imperfect collusion where cartel members partially internalize each other's behavior in a symmetric fashion.[23] Markups are given by

$$\mu_{sk}^{\mathcal{C}} = \left[\frac{\rho-1}{\rho} + \left(\frac{\eta-1}{\eta} - \frac{\rho-1}{\rho}\right)\left(\omega_{sk} + \kappa \sum_{j\in\mathcal{C}\setminus\{k\}} \omega_{sj}\right)\right]^{-1}. \tag{12}$$

This is a sales-weighted harmonic average of the within- and between- markups as in the benchmark case, except that the weight is augmented to reflect the market power of the other firms in the cartel. This effect is more pronounced as the collusion intensity κ increases.

C. How does Collusion Distort the Market Structure?

Cartel Overcharge. — Consider transitioning from the benchmark competitive equilibrium to a collusive equilibrium with a small collusive intensity $\Delta\kappa$. Variable $\hat{x}$ denotes the percentage deviation of variable x from the competitive equilibrium. The following proposition states the effect of collusion on the sectoral price index, using a first-order approximation.

PROPOSITION 1: *Starting from the competitive equilibrium, symmetric collusion increases the sectoral price index. The increase is larger i) the greater the collusion intensity, $\Delta\kappa$ and ii) the larger the market share controlled by the cartel, $\omega_{s\mathcal{C}}$. In particular, the sectoral price increase is*

$$\hat{P}_s = \frac{1}{\rho-1}\frac{1}{1-\sum_k \omega_{sk}\Upsilon_{sk}} \sum_{k\in\mathcal{C}} \Upsilon_{sk}\left(\omega_{s\mathcal{C}} - \omega_{sk}\right)\Delta\kappa, \tag{13}$$

where $\Upsilon_{sk} := \frac{\omega_{sk}(\rho-1)\left(\frac{1}{\eta}-\frac{1}{\rho}\right)\mu_{sk}}{1+\omega_{sk}(\rho-1)\left(\frac{1}{\eta}-\frac{1}{\rho}\right)\mu_{sk}} \in (0,1)$ *represents the elasticity of the firm's own price with respect to the sectoral price index and* $\omega_{s\mathcal{C}} := \sum_{k\in\mathcal{C}} \omega_{sk}$ *is the cartel's combined market share.*

This intuitive result illustrates that both the intensive and extensive margins of the cartel are at play. Equation 13 entails that both an increase in the collusion intensity $\Delta\kappa$ and a larger market share controlled by the cartel $\omega_{s\mathcal{C}}$ lead to a higher price index in the sector. The latter directly echoes theoretical findings on heterogeneous cartels.[24]

[23] This is the case considered by Brooks, Kaboski and Li (2021) who study how Chinese industrial clusters affect competition.

[24] See Theorem 3 in Bos and Harrington (2010).

PROPOSITION 2: *Starting from the competitive equilibrium, symmetric collusion increases the markups of all firms within the cartelized industry. In particular, i) for cartel members, the markup increase declines with firm size ii) while for non-cartel members, the markup increase increases with firm size. For cartel members ($k \in \mathcal{C}$) in sector s, the markup change is given by*

$$\hat{\mu}_{sk}^{\mathcal{C}} = \underbrace{\Upsilon_{sk}\hat{P}_s}_{\textit{Umbrella Pricing}} + \underbrace{\frac{1}{\rho-1}\frac{\Upsilon_{sk}}{\omega_{sk}}\left(\omega_{s\mathcal{C}} - \omega_{sk}^{\mathcal{C}}\right)\Delta\kappa}_{\textit{Cartel Overcharge}}. \tag{14}$$

The first term is common to all firms in the cartelized sector, whether they are part of the cartel or not. It can be interpreted as a form of umbrella pricing, reflecting changes in competition. The second term is specific to cartel members and can be interpreted as the cartel overcharge. The overcharge varies across firms in the cartel and is increasing along both the extensive margin of collusion, i.e. the total market share controlled by the cartel $\omega_{s\mathcal{C}}$ and the intensive margin $\Delta\kappa$. Under symmetric collusion, the overcharge tends to be larger for smaller cartel members.[25] Collusion increases markups for *all* firms in the cartelized industry compared to the competitive equilibrium, as it raises the sectoral price index, boosting demand for non-cartel members and enabling higher markups through an umbrella pricing effect.

Market Shares Reallocation: a Collusion Paradox?. — Cartel members charge higher markups than they would in a competitive equilibrium, resulting in a price increase which reallocates market shares towards competitive firms. In a cartelized industry, the evolution of market shares is characterized as follows.

COROLLARY 1: *Non-cartel members all gain market shares. Among cartel members, the evolution of market shares depends on the composition of the cartel. In particular:*

$$\begin{cases} \hat{\omega}_{sk} = (\rho-1)\left(1-\Upsilon_{sk}\right)\hat{P}_s, \\ \hat{\omega}_{sk}^{\mathcal{C}} = (\rho-1)\left(1-\Upsilon_{sk}\right)\hat{P}_s - \Upsilon_{sk}\left(\frac{\omega_{s\mathcal{C}}}{\omega_{sk}^{\mathcal{C}}} - 1\right)\Delta\kappa. \end{cases} \tag{15}$$

Non-colluding firms experience a price increase due to the umbrella pricing mechanism, though to a lesser extent than the overall sectoral price increase ($\Upsilon_{sk} < 1$). This allows them to gain market shares at the expense of the cartel. That some cartel members may lose market shares can appear surprising at first. In fact, it follows naturally from standard assumptions and is well in line with

[25] This feature follows directly from the simplifying assumption that the collusion intensity κ is the same across cartel members. Alternatively, one can consider arrangements with varying collusion intensity, or, parsimoniously, back-out the firm-specific collusion intensities that deliver the same overcharge for all cartel members, as derived in online Appendix D5.

empirical findings from the IO literature. From a theoretical standpoint Bovin and Bos (2023) show that cartel members generically lose market shares in a large class of oligopoly models under mild conditions.[26] This class encompasses models where the cartel operates as a price or quantity leader (d'Aspremont et al., 1983; Shaffer, 1995), in models with capacity constraints (Bos and Harrington, 2010), or in models with vertically differentiated products (Bos, Marini and Saulle, 2020). One notable exception is the case of dominant-firm cartel conduct when the cartel can directly restrict non-cartel firms' supply, for instance through coercion or political lobbying.

Empirically, losses of market shares for cartelized firms consistent with the theory have been documented in the truck industry (Bovin and Bos, 2023), in the German cement industry (Hschelrath and Veith, 2016), in the French inter-war steel cartels (Barbezat, 1996). More famously, Chinese competitors outside the cartels saw their market shares grow during the well-known global citric acid and vitamin C conspiracies (Connor, 1998).

D. Collusion, Productivity, and Welfare

How does collusion affect sectoral productivity z_s, aggregate productivity A, and welfare?

Sectoral Productivity. — Proposition 3 shows how collusion affects sectoral productivity.

PROPOSITION 3: *The impact of collusion on sectoral productivity is given by*

$$\hat{z}_s = \rho Cov_\omega \left[\frac{\mu_s}{\mu_{sk}}, \hat{\mu}_{sk} \right],$$

where μ_s is the sectoral markup, the sales-weighted average of x_k is denoted by $E_\omega[x_k] = \sum_k \omega_k x_k$ and the sales-weighted covariance of any two variables x_k and z_k writes $Cov_\omega[x_k, z_k] = E_\omega[x_k z_k] - E_\omega[x_k]E_\omega[z_k]$.

It turns out that the effect of cartels on sectoral productivity is theoretically ambiguous and depends on the composition of the cartel. Since firm-level productivity remains unchanged, sectoral productivity can only be affected by market share reallocations due to resource misallocation (Baqaee and Farhi, 2020). Intuitively, if the cartel is made up of top producers, their markup increase shifts demand towards less productive firms within the sector. As a result, the covariance term is negative and sectoral productivity decreases. Conversely, a cartel made up of the smallest firms in the sector would redirect demand towards larger, more productive firms, generating an increase in sectoral productivity.

[26] These assumptions are that 1) profit functions are twice differentiable, 2) strictly concave in own prices, 3) marginal profitability increases with the prices set by rivals, and 4) own price effects dominate the combined cross price effects. See Bovin and Bos (2023) for details.

Aggregate Productivity. — The model can be aggregated analytically, which yields a transparent analysis of the impact of distortions on productivity and welfare. In particular, output in this economy can be represented by an aggregate production function $Y = AL$, where A measures aggregate productivity and L is total labor employed in the economy. Aggregate productivity follows from the first-order condition for the optimal use of labor combined with the labor market clearing condition and is given by

$$A = \left[\int_0^1 \left(\frac{\mathcal{M}}{\mu_s}\right)^{\eta} z_s^{\eta-1} ds\right]^{\frac{1}{\eta-1}}, \tag{16}$$

where $\mathcal{M}$ is the aggregate markup level. Aggregate productivity is not at its first-best level because of markup dispersion arising from firm heterogeneity. More productive firms have more market power, produce less than what is socially optimal, which results in a suboptimal allocation. Indeed, the efficient productivity level A_{eff} is instead given by

$$A_{\text{eff}} = \left(\int_0^1 z_s^{\eta-1} ds\right)^{\frac{1}{\eta-1}}. \tag{17}$$

Our main exercise consists in comparing the aggregate productivity level obtained in the presence of cartels A_{cartel} to the one that would be obtained in the *competitive* Nash-Cournot equilibrium A_{comp}. Any difference between these two productivity levels arises from changes in markup dispersion. This is different from the exercise done by Edmond, Midrigan and Xu (2015) and Baqaee and Farhi (2020) who are instead interested in comparing A_{comp} to the efficient productivity level A_{eff} obtained in the *absence* of markup dispersion. We will also compare A_{cartel} to A_{eff} in our calibrated economy with collusion and A_{comp} to A_{eff} in a calibrated version of the model without collusion ($\kappa = 0$).

Aggregate Welfare. — For the model to generate changes in welfare, we consider the static formula developed in Edmond, Midrigan and Xu (2023).[27] In this case, the level of the aggregate markup acts as a distortionary wedge. Intuitively, an increase in the aggregate markup induced by the cartelization of the economy reduces the aggregate scale of production and decreases the representative consumer's welfare. In the model, the aggregate markup changes as cartels generate market share reallocations and all firms experience a markup increase within cartelized sectors (Proposition 2). We compute the welfare change in consumption-equivalent units as detailed in online Appendix D7.

[27] We favor this formula over a standard extension with capital accumulation and elastic labor supply as it is consistent with our framework with labor as the only factor of production and is faster to compute. Results remain qualitatively unchanged and quantitatively similar when using the extension with capital accumulation that accounts for transitional dynamics.

III. Quantification of the Model

We now turn to the description of our calibration strategy. The key parameters determining the extent to which aggregate productivity varies in the presence of cartels are the within and across-sector elasticities of substitution ρ and η, respectively, and the collusion intensity parameter κ. The extent to which cartel members internalize the effect of their decision on other cartel members depends on the parameter κ, which governs the amount of cartel overcharge. We first describe how we parameterize the model before discussing its fit.

A. Parameterization

We allow sectors to be ex ante heterogeneous, meaning that some sectors are initially more productive than others. With some abuse of terminology, we refer to the largest sectors as the most productive ones, as more productive sectors ex ante tend to be larger ex post, though the correlation is not perfect. Given Proposition 3, four extreme cases can be considered. First, cartelized industries are the largest, and cartels consist of the most productive firms. In that case, one can expect large *losses* from collusion. Second, cartelized industries are the least productive, while cartels consist of the most productive firms. In this case, one could expect small losses from collusion. Third, cartelized industries are the largest, but cartels consist of the least productive firms. In this case, one could even expect *gains* from collusion.[28] Finally, cartelized industries are the least productive, and cartels consist of the least productive firms. In that case, one could expect small gains from collusion. Since our goal is to quantify the static cost of cartels and understand the extent to which cartels can be costly, while abstracting from targeted entry, we must assign cartels to specific industries and determine their members. Given the evidence reported in Table 1 and Table 2, we assign cartels to the most productive industries and assume that cartels are made up of the most productive firms within the cartelized industries. This likely provides an upper bound on the static cost of cartels.

Assigned Parameters. — We set the inverse of the Frisch elasticity of labor supply to 2. This parameter is used to assess the effect of cartels on welfare. The value is reported in Table 4.

Productivity Distribution. — Firm k's productivity level z_{sk} is the product of a sector-specific component z_s and an idiosyncratic one a_{sk} (Edmond, Midrigan and Xu, 2015): $z_{sk} = z_s a_{sk}$. We assume that the productivity distribution is Pareto. Within each sector, firms draw a_{sk} from a Pareto distribution with

[28] This is true for sectoral productivity. Aggregate productivity, however, also depends on changes in sectoral markups as shown in equation (16).

shape parameter ξ_k, independently across firms. This parameter determines the amount of concentration within sectors. The sectoral productivity term z_s is also drawn from a Pareto distribution, with shape parameter ξ_s, which determines the amount of concentration across sectors. When ξ_k and ξ_s increase, the productivity distribution becomes less skewed and firms and sectors become more homogeneous —leading to lower concentration ratios, which we target below.

Share of Cartelized Industries. — There can only be one cartel per sector s. Not all sectors, however, are cartelized. We assume that 10% of industries are cartelized, corresponding to the average share of cartelized industries over our sample period (online Appendix Table E4). Given the secret nature of cartels in France and the resulting uncertainty regarding the share of cartelized industries, we will show how our results change depending on the share of cartelized industries.

Cartelized Industries. — Given the results in Table 1, we assume that cartelized industries are the most productive. Specifically, we allocate cartels to the top 10% of ex ante more productive sectors. While this approach helps match the positive correlation reported in Table 1, it does not necessarily capture its exact magnitude. We return to this point in Sections IV and V.

Number of firms per sector. — The number of firms per sector is drawn from a geometric distribution with parameter $\sigma \in (0, 1)$ so that the probability of having K_s firms is given by $\sigma(1-\sigma)^{K_s-1}$. The parameter σ pins down the number of firms per sector: for instance, $1/\sigma$ yields the average number of firms. The parameter σ therefore directly influences the number of firms operating in each sector; as σ increases, the number of firms decreases.

Number of Cartel Members. — The number of cartel members is drawn from a geometric distribution with parameter $\zeta \in (0, 1)$ so that the probability of having $K_{\mathcal{C}}$ members is given by $\zeta(1-\zeta)^{K_{\mathcal{C}}-1}$. The parameter ζ pins down the number of cartel members, as the average number of cartel members is $1/\zeta$. This implies that cartels may consist of two, three, four, or more members. The parameter ζ therefore directly governs the number of cartel members in cartelized sectors; as ζ increases, the median number of cartel members decreases.

Cartel Composition. — Table 2 and additional evidence detailed in online Appendix B justify our assumption that whenever a cartel is present in an industry, it is made up of the most productive firms. We return to this point in Section V.

B. Markup Estimation

We rely on the seminal work of De Loecker and Warzynski (2012) to recover firm-level markups. The main advantage of this method is that it does not require imposing strong parametric assumptions about market structure or demand. The formula for firm-level markups μ_{kt} is given by:

$$\mu_{kt} = \frac{\beta^m_{kt}}{\alpha^m_{kt}}, \tag{18}$$

where β^m_{kt} is the output elasticity of materials in year t and α^m_{kt} is that input's revenue share, i.e. expenditures on materials over firm-level turnover. Intuitively, markups drive a wedge between the marginal revenue product of any flexible input and that input's marginal cost. Two pieces of information are required to recover the distribution of firm-level markups: the output elasticity of a flexible input and the revenue share of that input. The latter is readily available in our data, whereas the former requires taking a stance on the production function. To estimate firm-level markups, we use a more general production function that includes not only labor but also capital, materials, and services.

As we do not have data on prices over our time period, we circumvent the issue recently raised by Bond et al. (2021) of estimating production functions without data on prices and further explored by De Ridder, Grassi and Morzenti (2025). We instead adopt the cost-share approach of Syverson (2004) and Foster, Haltiwanger and Syverson (2008), which assumes constant returns in production. To further address potential biases from non-neutral technological differences affecting output elasticities and markup estimates (Raval, 2023*b*), we use Raval (2023*a*)'s cost-share estimator. This estimator groups firms into bins, based on their labor-to-materials cost ratio, accounting for variations in labor-augmenting productivity. Output elasticities are recovered as input cost shares within each bin, using materials as the flexible input. Our output elasticities vary at the 4-digit industry-quintile level. To remove outliers, we winsorize firm-level markups at the top and bottom 5% level by calendar year. More details on the variables used and on the methodology can be found in online Appendix C.

As shown in online Appendix Table E1, our markup distribution is not highly skewed. The aggregate markup level, calculated as a sales-weighted harmonic average, is 1.14 in 2007, which is of a similar order of magnitude to the average value of 1.08 reported in De Ridder, Grassi and Morzenti (2025). Online Appendix Figure F6 reports the evolution of the aggregate markup based on our distribution of firm-level markups. While the level is not directly comparable to existing studies —since the underlying distribution of firm-level markups differs —the aggregate markup series we recover displays an upward trend over the period, consistent with the evidence reported in De Ridder (2024).

C. Collusion Intensity Parameter

The equilibrium relationship between inverse markups and own market shares and cartel market shares can be used to recover the intensity of collusion, as in Brooks, Kaboski and Li (2021).

In equilibrium, the model yields

$$\frac{1}{\mu_{sk}^{\mathcal{C}}} = \frac{\rho - 1}{\rho} - \left(\frac{1}{\eta} - \frac{1}{\rho}\right)(1-\kappa)\omega_{sk} - \left(\frac{1}{\eta} - \frac{1}{\rho}\right)\kappa \sum_{j \in \mathcal{C}} \omega_{sj}. \tag{19}$$

We can thus write the following regression

$$\frac{1}{\mu_{sk}^{\mathcal{C}}} = a_0 + a_1 \omega_{sk} + a_2 \sum_{j \in \mathcal{C}} \omega_{sj} + \nu_{sk}, \tag{20}$$

where $a_0 = \frac{\rho-1}{\rho}$, $a_1 := \frac{(1-\kappa)(\eta-\rho)}{\eta\rho}$ and $a_2 := \frac{\kappa(\eta-\rho)}{\eta\rho}$. The parameter κ can be recovered from the estimated parameters:

$$\hat{\kappa} = \frac{\hat{a}_2}{\hat{a}_1 + \hat{a}_2}. \tag{21}$$

Moreover, the sum of the two slope parameters yields a restriction on the value of the elasticities of substitution, allowing us to determine η given the value of ρ and vice-versa. Indeed,

$$\left(\frac{1}{\rho} - \frac{1}{\eta}\right) = \hat{a}_1 + \hat{a}_2. \tag{22}$$

We rely on cross-sectional variation to identify these reduced-form parameters as adding firm fixed effects would make us lose about a third of cartel members. We return to this point in Section V.[29]

The results are provided in Table 3. The first column reports the regression on the sample of colluding firms without controlling for their joint market share. Firms with a higher market share charge higher markups: the point estimate on a firm's own market share is equal to -0.27 and is significant at the 5% level. Column 2 shows that the cartel market share is also negative and significant at the 1%

[29] We note that the parameter ρ could, in principle, be inferred from equation (20). There are at least two reasons to instead calibrate it. First, this helps us build the bridge with previous work targeting similar moments (Atkeson and Burstein, 2008; Edmond, Midrigan and Xu, 2023). Second, in some instances, the constant term may yield a negative ρ parameter, which would be inconsistent with the theoretical restriction that $1 < \eta < \rho$. The observation that it might be hard to find parameter values consistent with this restriction and that can rationalize observed markup dispersion is further developed in Blaum, Lelarge and Peters (2018).

Table 3—Estimation of κ

Dependent variable:	$\frac{1}{\mu_{kt}}$			
	(1)	(2)	(3)	(4)
Firm's market share	-0.269		-0.158	-0.143
	(0.112)		(0.126)	(0.123)
Cartel's market share		-0.143	-0.093	-0.100
		(0.042)	(0.047)	(0.045)
Year FE	No	No	No	Yes
Observations	2,397	2,397	2,397	2,397
Adj. R^2	0.01	0.01	0.02	0.04

Notes: Standard errors clustered at the firm level. The collusion intensity is estimated following equation (20). The dependent variable is the firm's inverse markup. The cartel market share variable is the sum of the market shares of all firms that belong to the same cartel-4-digit industry pair.

level.[30] In column 3, we include both market share variables, as required by the model. The coefficient on a firm's own market share remains negative but is no longer significant. Consistent with the theoretical model, the joint market share coefficient is negative and significant at the 10% level. In column 4, we further include year fixed effects to control for time-varying unobserved heterogeneity common across cartel members. The point estimates and standard errors change very little. We find that these point estimates yield $\kappa = 0.41$. This estimate is larger than the value of $\kappa = 0.3$ found by Brooks, Kaboski and Li (2021). This is consistent with cartels detected and convicted by competition authorities having a higher collusion intensity than firms in special economic zones (SEZs) exhibiting cooperative pricing behavior in China.

In online Appendix Table E9, we show that the results remain robust when using firms' labor share or the ratio of sales to variable costs as markup measures, winsorizing the distribution at various levels, or defining market shares based solely on sales to France. The value of κ ranges from 0.29 to 0.83.

One should note that the estimating equation does not deliver parameters that can be given a causal interpretation. In the model, markups and market shares are jointly determined in equilibrium, which raises a simultaneity concern. Finding exogenous variation in market shares is not straightforward in our context, and we therefore attempt to mitigate this issue through an instrumental-variable strategy. Our goal is to assess whether our baseline estimate of κ is conservative.

[30]Cartel market shares are defined at the cartel-4-digit-industry-year level, as some cartels operate across multiple 4-digit industries. This definition ensures that the cartel market share is well-defined in the data.

We use pre-collusion own and cartel market shares —market shares measured in the first year in which cartel members appear in the sample —as instruments for current market shares. The idea is that these pre-cartel market shares are predetermined with respect to the markup shocks operating during the collusive period, thereby alleviating the concern that current shocks jointly drive both market shares and markups. The inclusion of year fixed effects further removes macroeconomic shocks common to all firms in a given year. Online Appendix Table E10 shows that the cartel-share coefficient remains negative in both columns, and the implied value of κ increases to around 0.6. We also investigate how our results change when adding different sets of fixed effects, such as sector or cartel fixed effects. The concern, however, is that market shares exhibit limited within-sector and within-cartel variation, so these fixed effects remove much of the identifying variation.[31] As shown in online Appendix Tables E11 and E12, including these fixed effects indeed results in noisier estimates. The coefficient on the cartel market share is no longer statistically significant at conventional levels but remains negative. If anything, the implied value of κ under these alternative specifications is larger than our baseline estimate, ranging from 0.6 to 1.65. While these estimates should be interpreted with caution, they suggest that our baseline estimate of κ is conservative.

Finally, note that the slope parameter can theoretically be recovered using the sample of non-cartel members. In online Appendix D8, we show that this restriction can be applied interchangeably to both cartel members and non-cartel members. However, for our baseline results, we choose to rely on the sample of cartel members, as competitive firms could potentially be undetected cartel members, affecting the point estimate of interest. Online Appendix Table E13 shows that the slope parameter for the sample of competitive firms is -0.16 and is significant at the 1% level. Finally, we conduct a placebo test by testing whether the cartel market share is negatively correlated with inverse markups during periods when cartel members do not collude. As expected, the point estimate for the cartel market share is negative and not statistically significant, as shown in online Appendix Table E14.

D. Calibrated Parameters

Our model has six parameters that need to be estimated:

$$\theta \equiv \left\{ \rho, \eta, \xi_s, \xi_k, \sigma, \zeta \right\}$$

[31] In the extreme case where a cartel operates in only one 4-digit industry, including cartel-year fixed effects would absorb all of the variation used to identify the cartel market-share coefficient. As noted above, cartel market shares vary at the cartel-4-digit-industry-year level, and for many cartel-year pairs cartel members span only a small number of 4-digit industries, leaving very limited within-cartel-year variation in cartel market shares.

where θ is a vector of model parameters. These parameters are chosen in order to minimize the following model-data distance function (Acemoglu et al., 2018):

$$\sum_{m=1}^{M} \frac{\left|\text{Moment}_m(\text{Data}) - \text{Moment}_m(\text{Model}, \theta)\right|}{\frac{1}{2}\Big(\left|\text{Moment}_m(\text{Data})\right| + \left|\text{Moment}_m(\text{Model}, \theta)\right|\Big)}$$

where m denotes each moment and M is the total number of moment targets. We now discuss the moments that help us identify our parameters.

Aggregate Markup. — To determine the elasticity of substitution across firms, we require our model to match a given aggregate markup value, $\mathcal{M}$. Since the aggregate markup is a sales-weighted harmonic average of firm-level markups, and small firms set markups in line with the elasticity of substitution within sectors, this demand elasticity has to be lower when the aggregate markup target is higher. We target a value of $\mathcal{M} = 1.14$, which we estimated for 2007.

Markups and Market Shares. — As detailed above, the estimated slope parameters $\hat{a}_1 + \hat{a}_2 = -0.24$ help identify η conditionnal on ρ. Indeed, from equation (22), $\eta = \frac{1}{\rho^{-1} - (\hat{a}_1 + \hat{a}_2)}$. Intuitively, the gap between ρ and η pins down how dispersion in market shares translates into markup dispersion.

Concentration. — To pin down the Pareto shape parameters ξ_s and ξ_k, we target several moments of the distribution of sales concentration across and within sectors in 2007. The top-4 and top-20 sales share within French industries are 41% and 60%, respectively —each industry is weighted by its sales share in the economy. Moreover, the top 4 (20) industries account for 14% (33%) of overall sales, which helps us identify ξ_s.

Median Number of Firms per Sector. — In our administrative data, the median number of firms per 4-digit industry is 178. We therefore target a median of 178 firms per sector.

Median Number of Cartel Members. — The median number of cartel members is four (online Appendix Table E2). We therefore target a median of four cartel members in our calibration. Individual cartels may have two, three, four, or more members, but the median number of members across all cartels should match this value.

Table 4—Baseline Calibration

Panel A. Targets			
Moments	*Data*	*Collusion*	*Competitive*
Aggregate markup	1.14	1.14	1.14
Slope parameter	-0.24	-0.24	-0.24
Top-4 sales share, within-sector	0.41	0.42	0.41
Top-20 sales share, within-sector	0.60	0.66	0.64
Top-4 sales share, across-sector	0.14	0.14	0.14
Top-20 sales share, across-sector	0.33	0.33	0.32
Median # firms per sector	178	178	166
Median # members per cartel	4	4	0
Panel B: Parameters			
Interpretation	*Parameter*	*Value*	*Value*
Substitution within sectors	ρ	12.06	11.09
Substitution between sectors	η	3.10	3.03
Pareto shape, firms	ξ_k	10.24	9.85
Pareto shape, sectors	ξ_s	2.81	2.79
Geometric, firms	σ	0.004	0.004
Geometric, cartel members	ζ	0.16	
Collusion intensity	κ	0.41	0
Inverse Frisch elasticity	ψ	2	2

Notes: Panel A reports the targeted moments generated by the model and their corresponding data counterparts. The parameters are chosen to minimize the distance between the model moments and the data moments, which are derived from French microdata. Panel B reports the parameter values used to generate the model moments.

E. Model Fit

Panel A of Table 4 reports the data and model moments, while Panel B display the parameter values.[32] Column 3 reports the calibration of our baseline model with collusion whereas column 4 displays the competitive version ($\kappa = 0$). We note that the elasticity of substitution within sectors is higher than that across sectors η, as required by the model. Overall, the model matches the aggregate markup, cartel overcharge, slope parameter extremely well, as well as the median number of cartel members and median number of firms per sector. Regarding the amount of concentration, the model generates a sales share for the top 20 firms within sectors a bit higher than in the data (0.66 instead of 0.6), while the sales share of the top 20 industries matches exactly its data counterpart. While the

[32]We have computed the Jacobian matrix of the baseline model's moments with respect to each estimated parameter evaluated at the calibrated value of the parameters (see online Appendix Figure F7).

Table 5—Untargeted Moments

Moments	Data	Model
Cartelized industry slope	0.20	0.61
Average cartel premium (sales)	2.44	3.35
Average cartel premium (employment)	2.57	3.28
Average cartel premium (labor productivity)	0.24	0.36
Average cartel premium (market share)	2.05	9.65
Standard deviation of log sales	1.37	1.35
Standard deviation of log employment	1.10	1.35
Average overcharge (in %)	10	4.72

Notes: The table presents untargeted moments for the model with collusion. The cartelized industry slope value comes from Table 1. Cartel premia are calculated within cartelized industries and then averaged across these industries. The standard deviations of log sales and log employment are computed over the entire sample. The average overcharge is computed by comparing firm-level markups of cartel members in the collusive equilibrium with those in the Nash-Cournot equilibrium, where $\kappa = 0$.

Pareto shape parameter ξ_k is close to the value reported in Edmond, Midrigan and Xu (2023) for an aggregate markup of 1.15, our elasticity of substitution across sectors is higher, although they target a slope parameter similar to ours. The difference arises because their model includes value-added, which effectively scales down their elasticity across sectors by the elasticity of output with respect to value-added.

Table 5 reports several untargeted moments. In the model, cartelized industries are slightly larger than in the data, and the model generates larger cartel premia within sectors. This is not surprising, given that we have assigned cartel members to the largest firms and industries. Moreover, we do not consider this to invalidate our model. Since we are dealing with secret arrangements, the cartel premia derived from the data may be underestimated if large cartel members are not detected by the competition authority. Our model accurately reproduces the standard deviations of log sales and log employment. Finally, the average overcharge generated by our model is close to 5%, compared to the literature's reported average cartel overcharges of 10%.[33] This discrepancy is due to the fact that our estimated intensity of collusion is relatively low, resulting in lower overcharges.

IV. How Costly Are Cartels?

We present the aggregate productivity and welfare gains from eliminating the competition distortions that arise because of cartels. We then explore alternative

[33]The OECD guide for helping competition authorities relies on a cartel overcharge of 10% (OECD, 2014). Moreover, Laborde (2021) finds that the median cartel overcharge is 10%.

Table 6—Aggregate Gains from Breaking Down Cartels

Calibrated model:	Collusion (1)	Competitive (2)
Panel A: Aggregate productivity gains, %		
$A_{\text{cartel}} \to A_{\text{comp}}$	0.21	
$A \to A_{\text{eff}}$	1.20	0.99
Distance to efficient allocation	-17.82	
Panel B: Aggregate welfare gains		
$\mathcal{M}_{\text{cartel}} \to \mathcal{M}_{\text{comp}}$ (in pp)	-0.82	
$\mathcal{W}_{\text{cartel}} \to \mathcal{W}_{\text{comp}}$ (in %)	0.45	
$\mathcal{W} \to \mathcal{W}_{\text{eff}}$ (in %)	5.41	5.21

Notes: The table displays the aggregate productivity gains (rows 1 to 3), the change (in points) in the level of the aggregate markup (row 4), and the change in welfare (rows 5 to 6) resulting from eliminating cartels. The figures are obtained by comparing the variables in the relevant equilibrium to that in the competitive Nash-Cournot equilibrium (rows 1, 4 and 5) and efficient allocation (rows 2, 3 and 6). The efficient allocation corresponds to the equilibrium without markup dispersion. The distance to the efficient allocation computed in row 3 is the ratio of the first two rows. Column 1 considers our calibrated model with collusion whereas column 2 considers a calibrated competitive oligopoly model. The corresponding parameter values are reported in Table 4.

quantitative exercises of interest.

A. Aggregate Productivity and Welfare Gains

Productivity. — Panel A of Table 6 displays the aggregate productivity gains of breaking down cartels or eliminating markup dispersion altogether starting from our calibrated baseline model (columns 1) or a calibrated competitive oligopoly model (column 2). We find that removing all cartels would increase aggregate productivity by 0.2%. Indeed, large firms charge lower markups and prices in the competitive equilibrium than in the cartel equilibrium. This contributes to redirecting demand towards top producers, thereby increasing aggregate productivity. The second row of the table computes the percentage difference between each allocation and its efficient counterpart —see equation (17). The misallocation cost of markups is 1.2% in the model with collusion. This is higher than what is reported in column 2 because our calibrated oligopoly model with collusion features *more* misallocation than its calibrated competitive counterpart. This shows that competitive oligopoly models, such as Atkeson and Burstein (2008) and more recently Edmond, Midrigan and Xu (2015, 2023), may substantially understate the misallocation cost of size-related markups when cartels are made up of top producers as is the case in our context. In fact, the productivity cost of markups computed with respect to the efficient frontier is 21% higher —from 0.99% in the

calibrated competitive economy to 1.2% in our calibrated baseline model —in the presence of cartels. The third row shows that removing all cartels could bring the economy up to 18% closer to the efficient allocation by eliminating the extra source of markup dispersion arising from anticompetitive behaviors.

Welfare. — The first row of Panel B of Table 6 investigates how removing cartels impacts the level of the aggregate markup. Eliminating all cartels would decrease the aggregate markup by about 0.8 points. When cartels are broken down, all firms in the cartelized industries start charging lower markups (see Proposition 2). Since larger former cartel members decrease their markup by a larger amount and have a larger market share, this has a large impact on sectoral indices and therefore on the aggregate markup.[34] Finally, the aggregate markup decrease has implications in terms of welfare.[35] The penultimate row shows that breaking down all cartels would lead to a consumption-equivalent welfare gain of 0.45%. Interestingly, the last row shows that eliminating markup dispersion altogether would increase welfare by 5.4% and 5.2% starting from the collusion and competitive models, respectively. The fact that these numbers are lower than those reported in the oligopoly version of Edmond, Midrigan and Xu (2023) is explained by the fact that both calibrated models initially feature *less* markup dispersion. Nonetheless, columns 1 and 2 show that the welfare gains from reaching the efficient allocation may be underestimated if collusion is not appropriately accounted for and cartels are made up of top producers.

Another important reference is the recent literature on the dynamic gains from higher markups, which suggests that higher markups can improve welfare through increased innovation (Cavenaile, Celik and Tian, 2019). In our model, cartel breakdowns do not lead to less innovation due to reduced profit opportunities. Although dynamic considerations could potentially outweigh static gains, this is not a foregone conclusion, even qualitatively. Indeed, the relationship between innovation and competition is complex and hump-shaped (Aghion et al., 2005). On the other hand, cartels may impose or reinforce barriers to entry, preventing productive firms from entering an industry or allowing low-productivity firms to enter industries with low barriers to entry (Carrera and Titov, 2019). Whether cartel breakdowns discourage innovation remains an open question and an important direction for future research.

B. Gains from Curbing the Collusion Intensity

In our main quantitative exercise, we have computed the aggregate productivity and welfare gains from *eliminating* cartels. In practice, this goal might be out

[34]Given that the aggregate labor share in the model is the inverse of the aggregate markup level, changes in the degree of cartelization of the economy further generate changes in the labor share due to changes in market concentration (Autor et al., 2020; De Loecker, Eeckhout and Unger, 2020).

[35]If the aggregate markup level were held constant, our static welfare formula would coincide with changes in aggregate productivity when comparing the cartel and competitive allocations.

Table 7—Aggregate Gains from Decreasing Collusion Intensity

	$\kappa \to 0.1$	$\kappa \to 0.2$	$\kappa \to 0.3$
	(1)	(2)	(3)
Panel A: Aggregate productivity gains, in %			
$A_{\text{cartel}} \to A_{\text{comp}}$	0.18	0.13	0.07
Distance to efficient allocation	-14.68	-10.58	-5.79
Panel B: Aggregate welfare gains			
$\mathcal{M}_{\text{cartel}} \to \mathcal{M}_{\text{comp}}$ (in pp)	-0.55	-0.33	-0.16
$\mathcal{W}_{\text{cartel}} \to \mathcal{W}_{\text{comp}}$ (in %)	0.34	0.22	0.12

Notes: The table displays the aggregate productivity gains (rows 1 to 2), the change (in points) in the level of the aggregate markup (row 3), and the change in welfare (row 4) resulting from decreasing the collusion intensity parameter κ from its baseline value of $\kappa = 0.41$. The efficient allocation corresponds to the equilibrium without markup dispersion. The corresponding parameter values are reported in Table 4.

of reach for antitrust authorities. We now show that antitrust enforcement can nevertheless achieve sizable gains by reducing the collusion *intensity* in the economy, instead of breaking down all cartels. When κ decreases but remains strictly positive, cartels are not dismantled but cartel members assign a lower weight to each other's profits. We thus think of a decrease in κ as a tougher antitrust environment making it harder for cartel members to sustain high markups.[36] Column 1 of Table 7 considers going from our benchmark value of $\kappa = 0.41$ to $\kappa = 0.1$, while columns 2-3 consider slightly higher weights equal to 0.2, 0.3, respectively. The aggregate productivity gains remain significant, ranging from 0.1% to 0.2%, and are decreasing in κ. The distance to the efficient allocation and the aggregate markup still decrease when cartel members assign a lower weight to each other but the effect is smaller in absolute value. The welfare gains range from 0.1% to 0.3%.

Our results point to the quantitative importance of the intensive margin of cartels. In this sense, a tougher antitrust environment that forces cartel members to decrease their supracompetitive markups yields aggregate gains.

C. Aggregate Cost of Umbrella Pricing

Following Proposition 2, *all* firms in a cartelized industry experience a decrease in their prices following the cartel breakdown. Some demand could thus be reallocated towards less productive non-cartel members, thereby dampening the effect of cartel breakdowns on aggregate productivity.

[36]Firms might be more reluctant to charge higher markups and prices if antitrust authorities can rely on antitrust tools to investigate anticompetitive practices. Specifically, the development of whistleblower tools or even the threat of increasing fines if customers complain or competition authorities start investigating might deter firms from maintaining the status quo in terms of anticompetitive pricing.

Online Appendix Table E15 examines the quantitative importance of the umbrella pricing effect. Specifically, the markups of non-cartel members are treated as exogenous primitives and remain fixed in both the cartelized and competitive economies. However, the markups of cartel members are allowed to decrease to their competitive equilibrium values. Column 2 shows that not accounting for the endogenous response of non-cartel members to increased competition from cartel breakdowns results in slightly higher aggregate productivity gains compared to the benchmark results in column 1. However, the effect on the aggregate markup is smaller in absolute terms because non-cartel members charge higher prices than they would if they could react to the change in competition. Despite this, the level of the aggregate markup remains important and leads to welfare gains equal to those found previously (0.45%).

In short, the endogenous response of non-cartel members attenuates the impact of cartel breakdowns on aggregate productivity and welfare but the effect is small.

D. Comparison with Harberger (1954)

Online Appendix Table E16 shows why our results differ from the classic estimate of Harberger (1954) who finds a dead-weight loss arising from monopoly distortions of 0.1% of GDP. Column 2 considers a version of the model in which industries contain a single firm charging a markup equal to the harmonic average of all the other firms' markups. This sectoral version of the model yields an estimate closer to that of Harberger (1954). We find that the aggregate productivity gains from eliminating cartels would be 0.06% without properly accounting for firm heterogeneity within sectors. Finally, to ease the comparison with Harberger (1954), we assume that the demand elasticities in the model are equal to unity.[37] The estimate displayed in column 3 shows that the aggregate productivity gain from breaking down cartels drops to 0.12%. This is because there are very little markup differences across firms when the elasticities of substitution are small, even in the presence of cartels. Eliminating them thus have a smaller quantitative effect.

Our results point to the importance of model misspecification, namely failure to properly account for heterogeneity within sectors and using appropriate demand elasticities, as also shown recently by Baqaee and Farhi (2020). While our framework abstracts from input-output linkages, it explicitly generates endogenous markups by modelling oligopolistic competition. Our results thus complement theirs by showing that the gains from eliminating total markups in their context or size-related supermarkups in ours can be higher than the estimate of Harberger (1954).

[37]The result displayed in the table relies on a version of the model in which aggregate consumption c is a Cobb-Douglas aggregator. In the absence of cartels, the price of a firm k is now given by $P_{sk} = \frac{\rho}{(\rho-1)-(\rho-1)\omega_{sk}} \times \frac{W}{z_{sk}}$. We then assume that ρ tends to unity. As an alternative, we also considered a version of our baseline model in which both η and ρ tend towards unity and $\rho > \eta$. We find that the gain from eliminating cartels in this case are also small.

Table 8—Robustness Experiments

	Decile 8 (1)	Top 20% (2)	$\kappa = 0.83$ (3)	Het. κ (4)	$\mathcal{M} = 1.1$ (5)	Bertrand (6)
$A_{\text{cartel}} \to A_{\text{comp}}$	0.06	0.24	0.44	0.22	0.24	0.06
$A_{\text{cartel}} \to A_{\text{eff}}$	1.08	1.19	1.38	1.11	1.28	0.34
Distance to efficient allocation	-5.61	-20.46	-31.68	-19.98	-19.08	-18.23
$\mathcal{M}_{\text{cartel}} \to \mathcal{M}_{\text{comp}}$ (in pp)	-0.20	-0.88	-0.96	-0.64	-0.64	-0.51
$\mathcal{W}_{\text{cartel}} \to \mathcal{W}_{\text{comp}}$ (in %)	0.12	0.50	0.72	0.41	0.44	0.21

Notes: The table displays the aggregate productivity gains (rows 1 to 3), the change (in points) in aggregate markups (row 4), and the change in welfare (row 5) resulting from eliminating cartels. The figures are obtained by comparing the relevant variables in the cartel equilibrium to that in the competitive Nash-Cournot equilibrium (rows 1, 4 and 5) and efficient allocation (row 2). The efficient allocation corresponds to the equilibrium without markup dispersion. The distance to the efficient allocation computed in row 3 is the ratio of the first two rows. Column 1 assigns cartels to sectors in the 8th productivity decile. Column 2 assigns cartels to the top 20% of sectors based on productivity. Column 3 recalibrates the model with a collusion intensity of 0.83. Column 4 allows the collusion intensity parameter to vary across cartels, with an average value of 0.41. Column 5 sets an aggregate markup target of 1.1. In column 6, firms compete à la Bertrand.

E. Robustness

In this section, we assess the sensitivity of our baseline results and recalibrate the model for each robustness experiment. The relevant model moments are reported in online Appendix Table E17.

Alternative Set of Cartelized Industries. — To study the extent to which cartels operating in the most productive industries matter, we assign cartels to sectors in the 8th productivity decile in column 1, or to the top 20% of sectors by productivity in column 2. In column 1, the aggregate productivity gain from removing collusion drops to 0.06% (from 0.21% in our baseline results) whereas the welfare gain drops to 0.12%. However, the industry premium is now underestimated (coefficient of 0.03 in the model instead of 0.2 in the data). In column 2, the aggregate gains are slightly larger than our baseline results. This highlights the significant impact that cartels operating in the most productive industries can have on aggregate productivity and welfare.

Alternative Collusion Intensity Value. — Our estimated gains are based on a collusion intensity value of $\kappa = 0.41$. Given the importance of this parameter in quantifying the gains of breaking down cartels, we recalibrate the model with $\kappa = 0.83$ in column 3, which is the highest value reported in online Appendix Table E9. This adjustment results in aggregate productivity and consumption-equivalent welfare gains that are approximately twice as large, at 0.44% and 0.72%, respectively. This version of the model also generates a larger average overcharge of 8%. In summary, our baseline estimates are conservative, as assuming a higher collusion intensity value leads to more substantial gains from breaking down cartels.

Heterogeneous κ. — In column 4, we allow κ to vary across cartels, i.e., we generate a distribution of intensity of collusion parameters $\kappa_{\mathcal{C}}$. Specifically, we model each cartel's intensity of collusion as a random draw from a uniform distribution over the unit interval. These draws are then rescaled so that the average matches our baseline value of $\kappa = 0.41$. Allowing κ to vary across cartels yields aggregate productivity gains from eliminating cartels averaging 0.22%, close to our benchmark estimate of 0.21%, whereas the welfare gains are 0.41%, in line with our baseline result of 0.45%. Overall, this provides reassuring evidence that allowing κ to vary across cartels yields results remarkably close to our baseline estimates.

Alternative Aggregate Markup Target. — Another important moment is the aggregate markup level, which is set equal to 1.14 for our baseline results. We consider an alternative value for this calibration target of 1.1. As shown in column 5 of Table 8, the productivity gains from breaking down cartels are equal to 0.24%. The consumption-equivalent welfare gains are 0.44%. Targeting an alternative aggregate markup value does not yield dramatic differences with our baseline estimates. Why is this the case? While ρ and η adjust to match this target, the relative gap between the two parameters remains unchanged. As a result of this, the amount of markup dispersion generated in the cartel allocation, and thus the gains from breaking down cartels, do not change dramatically either.

Bertrand Competition. — In our benchmark model, firms compete à la Cournot. We relax this assumption and instead assume that firms compete in prices in column 6.[38] As shown in online Appendix D1, the only point of departure from our benchmark model is that the demand elasticity is now a weighted arithmetic average of ρ and η. We find that the aggregate productivity gain is equal to 0.06% and is thus smaller than when firms compete in quantities. This is because the Bertrand model features less markup dispersion. This can also be seen from the fact that the cartelized economy is closer to the efficient allocation than before (row 2). Eliminating cartels reduces the distance to the efficient frontier by about 18%, however. The model nevertheless generates a large change in the *level* of the aggregate markup which decreases by 0.5 percentage points. This translates into a consumption-equivalent welfare gain of 0.2%.

V. Discussion

We discuss the cost of cartels operating in different industries in France in 1995 and 2007. We also discuss the composition of cartels and possible ways to extend

[38]Given that the Bertrand competition model does not generate equation (22), we instead set η to its baseline value, $\eta = 3.10$. The other parameters are chosen so as to match all the other moments. We have also experienced with setting $\eta = 1.01$: the results remain qualitatively and quantitatively similar.

Table 9—Gains from Breaking Down Cartels in 1995 and 2007

Year:	1995	2007
	(1)	(2)
Panel A: Aggregate productivity gains, %		
$A_{\text{cartel}} \to A_{\text{comp}}$	0.09	0.04
Distance to efficient allocation	-11.10	-4.30
Panel B: Aggregate welfare gains		
$\mathcal{M}_{\text{cartel}} \to \mathcal{M}_{\text{comp}}$ (in pp)	-0.18	-0.13
$\mathcal{W}_{\text{cartel}} \to \mathcal{W}_{\text{comp}}$ (in %)	0.15	0.08

Notes: The table displays the aggregate productivity gains (row 1), the change (in points) in the level of the aggregate markup (row 3), and the change in welfare (row 4) resulting from eliminating cartels. The figures are obtained by comparing the variables in the relevant equilibrium to that in the competitive Nash-Cournot equilibrium and efficient allocation (row 2). The corresponding parameter values are reported in online Appendix Table E17. In the model, cartels are assigned to sectors that have the same rank in terms of ex-ante sectoral productivity draws as the sectors in the data have in terms of sectoral market shares.

our framework. Finally, we also show that there may be other types of distortions in the data arising from common ownership.

A. Cartelized Industries in the Data

The numbers we have provided so far are likely on the upper end of the static cost of cartels, especially when κ is sufficiently large. We now assign cartels to industries as closely as possible to their operation in 1995 and 2007 —the years with the highest and lowest shares of cartelized industries over our time period, as shown in online Appendix Table E4 —and recalibrate the model (online Appendix Table E18). From the data, we determine the rank of the cartelized industries based on their market shares. We then assign cartels to sectors in the model with the exact same rank in terms of ex-ante sectoral draws. As before, note that while ex-ante more productive industries tend to be larger in terms of sectoral market shares, sectors are also ex-post heterogeneous due to the number of operating firms and idiosyncratic productivity draws. Nevertheless, this approach results in cartelized industries that are closer to the actual cartelized industries observed in the data for 1995 and 2007.

As shown in Table 9, the average cost of collusion is approximately 0.1% in terms of productivity and welfare. These numbers are smaller than our baseline result because, although some very large industries were cartelized in France during these years, they are not as large as in Section IV. Moreover, the industry premium is now 0.10 for 1995 and 0.12 for 2007 in the model, compared to 0.20 in the data (see Table 1).

This further stresses the significant quantitative impact of cartels operating in top industries. Given that some productive cartels may have operated in other large industries without being convicted, an aggregate cost of collusion of 0.1% likely represents a lower bound on the static cost of collusion.

B. Cartel Members in the Data

Our quantitative exercise assumes that detected cartels consist solely of the most productive firms, but it does not address how cartels form, whether it is rational for firms to join them, or how collusive arrangements are sustained over time. To better align with firm-level data, one may consider two approaches for generating flexible cartel arrangements. The first would be to endogenize the cartel formation choice. This could involve using a fixed cost parameter informed by relative size differences between cartel members. However, this approach would be computationally complex, as it would require evaluating numerous possible cartel compositions, given that firms influence industry-level outcomes. An alternative would be to implement the model on the data in the spirit of Brooks, Kaboski and Li (2021) and Di Giovanni, Levchenko and Mejean (2024). This approach would avoid parametric assumptions about productivity distributions or the number of operating firms. While this method would effectively capture the impact of detected cartels, our approach is better suited to highlighting the cost of cartels operating in specific industries, as explored in this paper.

Adopting alternative cartel arrangements could provide additional insights into the aggregate cost of collusion, making it an important area for future research.[39]

C. Alternative Types of Distortions

Since our microfoundation of collusive behaviors relies on cross-ownership structures, we test whether the screening equation (19) also applies to firms belonging to a group. The idea is that belonging to the same parent company could help firms coordinate their pricing strategies and generate distortions —see Brooks, Kaboski and Li (2021) for a similar application in China. To explore this issue, we use the French "Liaisons Financières" (LIFI) survey (Insee, 1994-2007*a*). This non-exhaustive survey contains information on the ownership status of firms headquartered in France. We estimate equation (19) on the sample of firms that belong to a group and compute the group market share as the sum of the market shares of firms owned by the same parent company within a 4-digit industry in a given year. Given the large sample size, we can also study whether firms coordinate their pricing strategies over time by including firm fixed effects. As shown in online Appendix Table E19 and online Appendix Table E20, where we

[39]We derive an expression for the profit incentives of the firms to join cartels and discuss how it fits with findings from the theoretical literature on cartel formation (Bos and Harrington, 2010) in the online Appendix. Online Appendix Figure F8 shows that all cartels experience aggregate profit gains from collusion.

exclude cartel members as a robustness check, we find that the coefficients on the overall group's market share are negative and statistically significant, while the coefficients on the own shares are not significant. This result holds with different sets of fixed effects, including firm fixed effects.

Overall, this test provides evidence that firms in France may coordinate their pricing strategies within ownership structures.

VI. Conclusion

We study the impact of collusion on aggregate productivity and welfare. We build an oligopolistic competition model with heterogeneous firms and show how collusion can amplify or dampen misallocation in otherwise standard competitive oligopoly models.

Calibrating our model to French data, we find that there are important static gains from breaking down cartels, especially if very large and productive sectors are cartelized. Cartels may reduce aggregate productivity by as much as 0.1% to 0.4% and may be as costly as 0.1% to 0.7% in terms of consumption-equivalent welfare. In terms of policy, our results suggest that antitrust enforcement and competition laws that aim to break down cartels can yield large gains at both extensive and intensive margins. Moreover, to ensure that cartels do not form and dampen productivity gains, other policies aiming to promote economic growth such as industrial policies or trade liberalization reforms may also be accompanied by robust competition laws.

There are several directions our work could take. First, collusion and misallocation are likely important issues in several emerging markets and developing economies. Second, our framework could be used to study the impact of horizontal M&As on aggregate productivity, by allowing for eventual cost synergies. Third, we focused on product-side distortions created by cartels, while several recent important court cases involve conspiracies involving firms agreeing not to compete on the labor market. Finally, incorporating input-output linkages would be an important undertaking to understand how cartelization may affect firms along the supply chain. These important questions are left for future research.

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